Oct 20, 2026 (Due: 08:00 Oct 27, 2026)

1. Let (λ, x) be a normalized eigenpair of an Hermitian matrix A. Suppose that $\tilde{x}$ is an approximate eigenvector that satisfies $\|\tilde{x} - x\|_2 = \epsilon \ll 1$. Show that

$$\left|\frac{\tilde{x}^* A \tilde{x}}{\tilde{x}^* \tilde{x}} - \lambda\right| = O(\epsilon^2).$$

2. Let P_{10} be the 10×10 Pascal matrix, whose (i, j)-entry is

$$\binom{i+j-2}{i-1} = \frac{(i+j-2)!}{(i-1)!\,(j-1)!}.$$

Compute the smallest eigenvalue of P_{10} using inverse iteration (INVIT) and Rayleigh quotient iteration (RQI). Visualize the convergence history.

3. Let

$$W_{10} = \begin{bmatrix} 10 & 1 & & & & & & \\ 1 & 9 & \ddots & & & & & \\ & \ddots & \ddots & 1 & & & & \\ & & 1 & 0 & 1 & & & \\ & & & 1 & \ddots & \ddots & & \\ & & & & \ddots & 9 & 1 \\ & & & & & 1 & 10 \end{bmatrix}$$

be the 21×21 *Wilkinson matrix*. Compute the five largest eigenvalue of W_{10} using subspace iteration. Try with 5-column and 6-column initial guesses. What do you observe?

4. Let $A_0 \in \mathbb{C}^{n \times n}$, μ_0, μ_1, …, $\mu_m \in \mathbb{C}$. Define A_1, A_2, …, A_{m+1} by

$$A_k - \mu_k I = Q_k R_k, \qquad A_{k+1} = R_k Q_k + \mu_k I,$$

for $k \in \{0, 1, \ldots, m\}$, where Q_k's are unitary matrices. Show that

$$(A_0 - \mu_0 I)(A_0 - \mu_1 I) \cdots (A_0 - \mu_m I) = (Q_0 Q_1 \cdots Q_m)(R_m \cdots R_1 R_0).$$

5. Implement the following algorithms for Hessenberg reduction:
(a) using Householder reflections;
(b) using Arnoldi process based on modified Gram–Schmidt orthogonalization.
Randomly generate a few matrices and compute the corresponding Hessenberg decomposition $A = QHQ^*$. Check the accuracy in terms of $\|Q^*AQ - H\|_{\mathsf{F}}$ and $\|Q^*Q - I\|_{\mathsf{F}}$ for your Hessenberg reduction implementations. What do you observe?

(optional) Perturb the matrix A a little bit. How do Q and H change accordingly?

6. (optional) Implement the plain QR algorithm and visualize the *componentwise* convergence on a small example.